
Cold Attack on the Mixed Quadrant of the $\forall PO$ -Free Fragment of $\mathcal{ALCI}_{\text{RCC5}}$

Finite cross-kernel stabilization, a class-top counterexample,
and a revised extraction program

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Principal result. For every fixed finite family of ascending or descending model towers, arbitrarily late finite recurrence segments can be selected so that every cross-kernel phase rectangle is constant. This is precisely the property needed by the cross-kernel universal obligation `kq_all`. The stronger assertion that entire cofinal tails can be made constant is false and is not required.

Abstract

The supplied attack packet isolates three open extraction problems for the mixed $\exists PO/\exists PP$ quadrant of the $\forall PO$ -free fragment of $\mathcal{ALCT}_{RCC5}$. This report makes progress on all three. First, it gives a finite simultaneous-rectangle theorem for finitely many interacting PP/PPI towers. Under the segment hypotheses stated in the packet, this proves the mathematical core of Target B at paper level. Second, it proves that the proposed Target A label is always satisfiable but constructs an eight-point ordered-disjoint countermodel showing that a common fresh class-top need not be insertable. Blocking by model type alone is therefore insufficient; a finite lower-context spectrum is a plausible replacement. Third, it refutes the natural tail-stabilize/add-witness fixed point for Target C and factors the remaining problem into postselection of uniform anchors followed by an ordered-disjoint PO-completion. The report also records several defects in the supplied Python probes. All conclusions are scoped to the abstract RCC5 composition-table semantics and the stated $\forall PO$ -free fragment. No complete mixed-quadrant extraction or decidability theorem is claimed.

Contents

1	Scope, assumptions, and result map	1
1.1	Evidence levels used in this report	1
2	Target B: finite simultaneous cross-kernel rectangles	2
2.1	The finite rectangle theorem	2
2.2	The transition ranks	2
2.3	Two towers: stabilization of the row limits	3
2.4	Finite fusion	3
2.5	Discharging the certificate obligations	3
2.6	Why infinite rectangular stabilization is false	4
2.7	Diagonal qualification	4
3	Target A: the class-top proposal fails	4
3.1	The local satisfiability lemma	4
3.2	An eight-point counterexample	4
3.3	No common fresh top	5
3.4	A bounded context refinement	6
4	Target C: the selection circularity is distinct	6
4.1	A counterexample to the finite fixed point	7
4.2	Postselection and anchored PO-completion	7
4.3	Dependence on Target B	8
5	Audit of the supplied probes	8

5.1	The Q3 upper diagnostic is tautological	8
5.2	The witness-selection cycle counts are unsound	8
5.3	The wp94 acceptance verdict is invalid	8
5.4	The wp93 proof-plan check is too weak	9
6	Revised work program	9
7	Conclusion	9
A	Executable artifacts in the accompanying package	10
B	Source basis and formal-status note	10

1 Scope, assumptions, and result map

We retain the packet’s standing assumptions without alteration:

- Models are total, converse-closed, composition-closed atomic RCC5 networks under abstract composition-table semantics; they need not be concrete region models.
- Strong equality is used: EQ is identity and occurs only on the diagonal.
- $x PP y$ means that x is a proper part of y ; PPI is its converse.
- No formula $\forall PO.C$ occurs in the closure. Thus a declared PO edge carries no universal obligation.
- The singleton cells used repeatedly are

$$\begin{aligned} \text{comp}(PP, PP) &= \{PP\}, & \text{comp}(PP, DR) &= \{DR\}, \\ \text{comp}(DR, PPI) &= \{DR\}, & \text{comp}(PPI, PPI) &= \{PPI\}. \end{aligned}$$

- Ordered-disjoint frames are used as the RCC5 normal form: a strict partial order, a symmetric irreflexive downward-closed disjointness relation, and PO as the residual relation.

The conclusions are as follows.

Target	Finding	Defensible status
A: group problem	The proposed root label is trivially satisfiable, but one common fresh class-top per (T, D) is refuted by an eight-point counterexample. A finite down-spectrum refinement gives a local gluing lemma.	Proposed repair refuted; bounded refinement open globally.
B: cross-kernel rect-angle	Finite recurrence segments admit simultaneous constant cross-kernel rectangles by a two-level stabilization and finite fusion argument.	Paper-level proof, pending Lean formalization and a diagonal guard check.
C: selection order	Tail stabilization followed by witness addition has no finite fixed point in general. Uniform witnesses can instead be postselected and PO-completed.	Geometric circularity separated; row-conservative demand closure remains.

1.1 Evidence levels used in this report

To avoid conflating proof, certification, and measurement, conclusions are labelled by their actual evidence level:

- **Paper-level theorem or counterexample:** a general mathematical argument is supplied here, but it has not yet been translated into Lean.
- **Finite executable check:** a dependency-free Python artifact checks a concrete frame, composition-table calculation, or bounded family. It is not a proof of the corresponding unbounded statement.

- **Reported certified background:** the input packet says a result is Lean-certified. The Lean development itself was not included in this package and was not independently recompiled for this report.
- **Candidate reduction:** a bounded construction or invariant has a local argument but still lacks one or more global certificate obligations.

2 Target B: finite simultaneous cross-kernel rectangles

Result: under the packet's fixed-family and recurrence hypotheses, the finite off-diagonal phase rectangles needed by the certificate have a paper-level proof. Infinite tail-by-tail stabilization is false and unnecessary.

2.1 The finite rectangle theorem

Let $K = \{0, \dots, m-1\}$ be finite. For each $k \in K$, let

$$c_k : \mathbb{N} \longrightarrow M$$

be a strict tower in one atomic RCC5 model. Its direction d_k is either PP or PPI:

$$R(c_k(n), c_k(n+1)) = d_k.$$

Assume, as in the packet's periodic extraction, that every tail contains two positions of the same full model type.

Theorem 2.1 (finite simultaneous rectangles). There are indices $s_k < t_k$ and atoms $q_{k\ell}$ for $k \neq \ell$ such that

$$\text{nty}(c_k(s_k)) = \text{nty}(c_k(t_k))$$

and

$$R(c_k(s_k + a), c_\ell(s_\ell + b)) = q_{k\ell}$$

for every $a < t_k - s_k$, every $b < t_\ell - s_\ell$, and every $k \neq \ell$. The intervals may be chosen beyond any prescribed finite collection of lower bounds.

Thus contiguous recurrence segments suffice; no arbitrary subsequence selection is used.

2.2 The transition ranks

Fix a point y and move x_n along a strict tower. If $R(x_n, x_{n+1}) = d$, composition requires

$$R(x_n, y) \in \text{comp}(d, R(x_{n+1}, y)).$$

For $d = \text{PP}$, the allowed transition graph has the rank partition

$$\{\text{DR}, \text{PP}\} < \{\text{PO}, \text{EQ}\} < \{\text{PPI}\};$$

for $d = \text{PPI}$, it has

$$\{\text{PPI}\} < \{\text{PO}, \text{EQ}\} < \{\text{DR}, \text{PP}\}.$$

Direct inspection of the RCC5 table shows that every allowed non-loop transition strictly increases the displayed rank. There are no transitions between distinct atoms at the same rank. Hence every relation sequence against a fixed point eventually stabilizes, with at most two genuine atom changes. The analogous target-coordinate statement follows by converse, or by using the corresponding right-composition transition relation.

2.3 Two towers: stabilization of the row limits

For two towers c and d , set

$$R_{ij} = R(c(i), d(j)).$$

One-sided stabilization gives, for each i , an eventual row value

$$\lambda(i) = \lim_{j \rightarrow \infty} R_{ij}.$$

Choose a target index beyond the stabilization thresholds for both rows i and $i + 1$. Applying composition to the triple $c(i), c(i + 1), d(j)$ shows that $\lambda(i), \lambda(i + 1)$ obey the same transition graph as a relation to a fixed point. The rank argument therefore yields

$$\exists I, q \forall i \geq I : \lambda(i) = q.$$

Choose any finite recurrence segment of c beyond I . Each of its finitely many rows has eventual value q , so moving d beyond the maximum of their individual thresholds makes the entire rectangle constant. A same-type recurrence interval for d can then be chosen inside that tail.

2.4 Finite fusion

Order the finitely many towers. When selecting tower k , place its complete recurrence segment beyond:

1. all outer row-limit stabilization bounds against towers not yet selected;
2. all row thresholds imposed by the finitely many phases already selected;
3. all stabilization thresholds imposed by the finite external set already fixed.

Finite type recurrence supplies equal-type endpoints after this maximum. Later towers are only restricted to farther tails, so previously obtained rectangles cannot be destroyed. Induction over the finite tower list proves Theorem 2.1.

2.5 Discharging the certificate obligations

Define

$$\text{phase}(k, a) = \text{mty}(c_k(s_k + a)), \quad Q(k, \ell) = q_{k\ell}.$$

If $\forall r. C$ belongs to a source phase and $Q(k, \ell) = r$, rectangular constancy gives relation r from that source point to every target phase point. Model semantics therefore places C in every target phase, exactly as required by **kq_all**.

Choose one representative from each selected segment. Because every macro relation is the actual relation between representatives, the finite macro-network is an induced subnet of the source model. Converse and composition closure, and hence the cross-kernel part of **frame_q**, follow immediately.

For distinct kernels the common rectangle atom cannot be **EQ** if recurrence gaps are chosen at least two: each segment then contains two distinct points of a strict tower, while a constant **EQ** rectangle would identify every point in both segments. Such recurrence gaps are available because some model type occurs infinitely often in every tail. This preserves strong equality at the macro level; coincident kernel traces may alternatively be merged before defining Q .

Rectangular constancy is sufficient, not logically necessary. Labels may satisfy **kq_all** accidentally, and $Q(k, \ell) = \text{PO}$ makes the implication vacuous in this fragment.

2.6 Why infinite rectangular stabilization is false

Let

$$M = \mathbb{Z} \times \{0, 1\}, \quad (n, u) < (m, v) \iff n < m,$$

with empty disjointness. The ordered-disjoint network has

$$R((i, 0), (j, 1)) = \begin{cases} \text{PP}, & i < j, \\ \text{PO}, & i = j, \\ \text{PPI}, & i > j. \end{cases}$$

Both coordinate sequences are ascending towers. Each individual row and column stabilizes, but every product of two infinite cofinal subsequences contains both PP and PPI. Finite constant rectangles occur arbitrarily late: select one finite segment completely above or below the other. This separates the true finite theorem from an invalid tail theorem.

2.7 Diagonal qualification

The displayed definition of `kq_all` in the packet lacks an explicit $k \neq k'$ guard. If this is literal, `frame_q` gives $Q(k, k) = \text{EQ}$, and `kq_all` would propagate the body of every $\forall \text{EQ}.C$ from one phase to every phase. That is stronger than equality semantics and stronger than the separately listed `kk_eq` obligation. The Lean definition should therefore be checked before the finite rectangle theorem is installed: either the guard is present, or the diagonal obligation must be changed explicitly.

3 Target A: the class-top proposal fails

Result: the advertised class-top label is always satisfiable, but this does not permit one fresh top to serve all same-type occurrences. A finite ordered-disjoint counterexample refutes that inference.

3.1 The local satisfiability lemma

Let $T = \text{mtv}_M(a)$, let $\exists \text{PP}.D \in T$, and define

$$L_T = D \sqcap \bigwedge \{E : \forall \text{PP}.E \in T\}.$$

Choose any w with $a \text{PP } w$ and $M, w \models D$. Every $\forall \text{PP}.E \in T$ is true at a , so $M, w \models E$. Consequently

$$M, w \models L_T.$$

This argument is valid in arbitrary relational semantics; the RCC5 table and the $\forall PO$ -free restriction are not needed. The proposed “is the label satisfiable?” question is therefore settled positively but is too weak to solve the group problem.

3.2 An eight-point counterexample

Use concept names P, Q, R, S and define

$$U_P = \forall \text{DR}.P, \quad U_Q = \forall \text{DR}.Q,$$

$$A = U_P \sqcup U_Q, \quad D = \exists DR.A.$$

Let C_0 be the conjunction of

$$\exists PP.D, \quad \exists PO.R, \quad \exists PO.S, \quad \exists PPI.\top, \quad P \sqcup \neg P, \quad Q \sqcup \neg Q.$$

This concept is in the mixed quadrant and contains no $\forall PO$ formula.

Take the eight points

$$x_1, v_1, y_1, z_1, x_2, v_2, y_2, z_2.$$

The strict order is exactly the two chains $x_i < v_i < y_i$, including their transitive edges. Disjointness holds exactly between z_i and the three points x_i, v_i, y_i . Every other distinct incomparable, non-disjoint pair is PO . This is an ordered-disjoint frame and therefore induces a composition-closed RCC5 network.

Use the following valuation.

Point	True atoms	Point	True atoms
x_1	P	x_2	Q, S
v_1	P, Q, R	v_2	P, Q, R
y_1	P, Q	y_2	P, Q
z_1	S	z_2	none

The DR-neighbours of z_1 are exactly x_1, v_1, y_1 , all of which satisfy P ; thus $z_1 \models U_P$. Since $x_1 \not\models Q$, $z_1 \not\models U_Q$. Symmetrically, $z_2 \models U_Q$ and $z_2 \not\models U_P$. Hence $y_i \models D$, witnessed by z_i , and $v_i \models \exists PP.D$, witnessed by y_i .

A direct evaluation of all 22 syntactic subformulas of C_0 gives

$$\text{mtv}(v_1) = \text{mtv}(v_2) = T.$$

The modal conjuncts at root v_1 also force the relevant lower contexts into the demand closure:

Demand at v_1	Unique witness
$\exists PP.D$	y_1
$\exists PO.R$	v_2
$\exists PO.S$	x_2
$\exists PPI.\top$	x_1

Thus v_2 is a genuine blocked same-type occurrence, and x_1, x_2 are not irrelevant ambient points. There is no positive $\forall PP$ formula in T , so $L_T = D$, already realized separately at y_1 and y_2 .

3.3 No common fresh top

Assume that an extension contains a fresh t with

$$v_1 PP t, \quad v_2 PP t, \quad t \models D.$$

Choose z such that $z DR t$ and $z \models A$. Since $x_i PP v_i PP t$, transitivity gives $t PPI x_i$. The singleton cell

$$\text{comp}(DR, PPI) = \{DR\}$$

therefore forces $z \text{ DR } x_1$ and $z \text{ DR } x_2$.

Boolean coherence at z chooses one disjunct of A :

- If $z \models \forall \text{DR}.P$, then $P(x_2)$ is forced, contradicting $x_2 \models \neg P$.
- If $z \models \forall \text{DR}.Q$, then $Q(x_1)$ is forced, contradicting $x_1 \models \neg Q$.

No common D -top exists in any label-preserving model or certificate extension. The same argument applies if the DR witness for t is represented by a kernel phase. Choosing the full type of one existing witness does not help: the obstruction is exposed while serving the nested demand D .

3.4 A bounded context refinement

For a source point v , define its strict lower type spectrum

$$\Delta_{<}(v) = \{\text{mty}(x) : x \text{ PP } v\}$$

and block by

$$q(v) = (\text{mty}(v), \Delta_{<}(v)).$$

If N is the number of full model types, at most $N2^N$ such contexts exist. With at most $m \leq |\text{cl}(C_0)|$ demands per context, the gate-child budget is at most $mN2^N$, independent of occurrence count and model depth.

Equality of lower spectra gives a local gluing lemma. Let a group have common context (T, Δ) , choose a representative PP witness, and take a disjoint copy of a witness-closed source gadget rooted there. Attach the copied upper cone above every group down-cone, propagate copied disjointness through the forced $\text{PP} \circ \text{DR}$ cell, and assign PO to all residual cross pairs. Every lower point has a type already present below the representative, so all forced PP and DR universal-compatibility rectangles already hold in the source model. Residual PO pairs carry no universal obligations.

This proves local frame preservation, cross-interface universal propagation, and internal existential fulfilment when the copied gadget is witness-closed. It does not yet give a finite certificate: the copied gadget may be infinite, and recursive context transitions can cycle. A plausible next step is to compile acyclic context states as externals and strict- PP strongly connected components as periodic kernels. The remaining proof must establish uniform PPI/DR witness rows and the cross-kernel obligations. Target A should therefore be recorded as a bounded reduction, not as solved.

4 Target C: the selection circularity is distinct

Result: the natural tail-stabilize/add-witness iteration can diverge even with one kernel. The certificate only needs constancy on selected phases, permitting postselection of uniform witnesses and an anchored PO -completion.

4.1 A counterexample to the finite fixed point

Use the concrete set-induced model

$$A_n = \{0, \dots, n\}, \quad B_m = \{m + 1\}, \quad n, m \in \mathbb{N}.$$

Then $A_n \text{ PP } A_{n+1}$ and

$$R(A_n, B_m) = \begin{cases} \text{DR}, & n \leq m, \\ \text{PPI}, & n \geq m + 1. \end{cases}$$

Let

$$H = (\exists \text{PP}.\top) \sqcap (\exists \text{DR}.\top), \quad C_0 = H \sqcap \forall \text{PP}.H.$$

Every A_n satisfies C_0 . A DR witness for A_n is a B_m with $m \geq n$. For fixed B_m , however, its row along the tower stabilizes only at $m + 1$, after changing from DR to PPI. Hence no n, m satisfy both

$$n \geq m + 1 \quad \text{and} \quad m \geq n.$$

The iteration “select beyond current external thresholds, add the selected segment’s witnesses, and reselect” therefore chases

$$n < m < n' < m' < \dots$$

and has no finite fixed point. This is not a counterexample to finite certificates. A postselected B_m can be constant on a one-phase segment and can be assigned a constant abstract row in the unfolding. The counterexample only refutes the unnecessarily strong eventual-tail extraction discipline.

4.2 Postselection and anchored PO-completion

Suppose a kernel segment $p_0, \dots, p_{q-1}$ has already been selected. If a certified uniform witness w satisfies

$$R(p_a, w) = r \quad \text{for every } a < q,$$

then w may be inserted after segment selection: this finite equality is exactly what the certificate function K requires. No stabilization against the unused tail is needed.

The following anchored PO-completion is a candidate local reduction. Partition uniform anchors by their required kernel relation into U_0 (PP), L_0 (PPI), and D_0 (DR). In the external frame put

$$\begin{aligned} U &= \uparrow U_0, & L &= \downarrow L_0, \\ D &= \downarrow D_0 \cup \{x : \exists u \in U, x \text{ DR } u\}. \end{aligned}$$

Assign the macro-kernel relation PP on U , PPI on L , DR on D , and PO on the residual points. Joint realization by any selected phase makes these sets compatible. The four singleton cells

$$\text{PP} \circ \text{PP}, \quad \text{PPI} \circ \text{PPI}, \quad \text{PP} \circ \text{DR}, \quad \text{DR} \circ \text{PPI}$$

show that every forced non-PO assignment agrees with the source model across all phases, provided the anchors and their closure remain row-conservative. Full model-type labels then discharge the corresponding `ke_all` and `ek_all` implications; residual PO rows are universally inert.

The unresolved issue is demand closure of the new anchors. If that closure remains model-agreeing, the anchored completion applies recursively. Borrowed, non-model-agreeing edges from Target A may violate the row-conservative invariant. The residue of C is thus a finite closure problem closely related to A, rather than another simultaneous tower stabilization theorem.

4.3 Dependence on Target B

Target B does not imply Target C: the preceding counterexample has one kernel, so the cross-kernel rectangle problem is vacuous. Conversely, postselection of external witnesses says nothing about Q or $\mathbf{kq_all}$. A clean extraction order is therefore:

$$\boxed{\text{select a finite kernel macro-frame using Target B}} + \boxed{\text{postselect uniform anchors and PO-complete using Target C}}.$$

The remaining common dependency is a bounded, row-conservative demand closure.

5 Audit of the supplied probes

The fixed-seed outputs were reproduced with Python 3. The empirical Q2 total in **wp131** is

$$1562/1702 = 91.77\%,$$

leaving 140 failing groups. The negative group phenomenon is robust, but several reported diagnostics or verdicts are incorrect.

5.1 The Q3 upper diagnostic is tautological

Q2 asks whether some source-model point y is above the gate and every blocked member and also satisfies D . Q3’s “has upper” predicate repeats the same condition after Q2 has failed. Its reported 0/140 is therefore forced by program structure. If the $D(y)$ test is removed, 44 of the 140 failures have a common order-upper and 96 do not. The README’s unqualified statement that none has “any common upper at all” is false.

Only 12 of the 140 failures have a nonempty set of positive $\forall PP$ bodies, whereas 102 have a modal filler D . The sample primarily stresses completion of the new top’s own demands, not aggregation of immediate PP -universal bodies.

5.2 The witness-selection cycle counts are unsound

Parts C and F of **wp131** reselect a target separately for each blocked borrower after gate expansion. Some selected targets are not members of **nodes**; **declared_cycle** silently ignores those edges. Some (a, D) groups also receive multiple targets even though the construction has one actual gate child. Redirecting all borrowers to that actual child yields eight cyclic instances under both repeat-free and “agree” selection, rather than five and zero. The five originally reported repeat-free survivors themselves are genuine; the claim that the agree rule removes all cycles is not.

5.3 The **wp94** acceptance verdict is invalid

The checker suppresses **ee_all** failures when a target filler is omitted by its budget truncation. Removing this exemption produces 33 violations in 29 of 3833 accepted instances. Moreover, nodes are pairs $(\mathbf{region}, \mathbf{budget})$ while the relation function ignores the budget. Distinct occurrences of the same region therefore receive off-diagonal EQ in 493 of 3833 instances,

including occurrences with different labels. This violates strong equality. The result is consistent with the packet’s later acknowledgement that the budget-truncated route was refuted, but `run_probes.sh` still prints a misleading “ALL PASS” verdict.

5.4 The wp93 proof-plan check is too weak

Part C returns success merely when its set of used cells is nonempty. Nine non-singleton composition cells are restricted rather than universal and still require arguments. Part B exhausts ordered-disjoint structures only through four points; it is useful finite evidence, not by itself a universal proof. The general result should rest on the existing formal ordered-disjoint normal-form proof, not on this finite enumeration.

6 Revised work program

The following order maximizes the chance of turning the present findings into a certified extraction theorem.

1. **Formalize the Target B transition ranks.** Prove that every non-loop left- and right-composition transition along PP/PPI strictly increases a three-valued rank.
2. **Formalize row-limit stabilization.** Apply one-sided stabilization twice: first to each row, then to the sequence of eventual row values.
3. **Formalize finite fusion.** Select equal-type recurrence segments in a finite kernel order and derive constant cross-phase rectangles, `kq_all`, and the induced macro-frame.
4. **Install the Target A counterexample as a regression.** It should reject any theorem claiming that type-only class tops always admit witness closure.
5. **Replace type-only blocking by strict down-spectrum contexts.** Prove the local gluing and compatibility-rectangle lemmas, then analyze the finite context transition graph.
6. **Compile context SCCs into kernels.** The outstanding obligations are uniform PPI/DR witnesses, `k_ex`, and interaction with the Target B macro-frame.
7. **State C as row-conservative postselection.** Avoid the refuted tail-fixed-point formulation; prove anchored PO-completion for a demand-closed uniform witness set.
8. **Repair the probes.** Separate source upper existence from D -carrying upper existence, retain actual gate children in cycle checks, enforce strong equality, and remove budget exemptions from obligation checks.

7 Conclusion

The packet’s recommended Target A lead is not the solution: its local satisfiability claim is immediate, and the inference to a shareable class-top is false. The counterexample shows exactly what a type-only key forgets: modal demands of the new top can inspect the union of the lower contexts through forced DR rectangles.

Target B, by contrast, has a short structural paper proof once the requested property is matched to the finite certificate. RCC5’s finite transition ranks support a second stabilization

over row limits, and a triangular finite fusion selects constant phase rectangles for all kernels simultaneously. Subject to the stated segment assumptions and the diagonal guard check, this removes the mathematical obstruction behind `kq_all`.

Target C is not another instance of B. Eventual stabilization against newly generated witnesses can diverge, but the certificate does not require it. Postselection of already uniform finite anchors, followed by ordered-disjoint PO-completion, isolates the remaining problem as row-conservative finite demand closure. That residue aligns naturally with the down-spectrum refinement suggested by Target A.

Accordingly, the overall mixed-quadrant extraction and the associated fragment decidability result remain open. The present report closes one finite combinatorial subproblem, refutes two proposed extraction disciplines, and narrows the remaining work.

A Executable artifacts in the accompanying package

The ZIP package contains three new dependency-free Python checks.

target_a_class_top_counterexample.py. Re-derives the RCC5 table from finite set semantics; checks the eight-point ordered-disjoint frame, closure-type equality, demand witnesses, and the forced-cell contradiction.

target_b_finite_rectangle_lemma.py. Checks the RCC5 transition ranks used in the proof and instantiates the interleaved-tower counterexample to infinite tail rectangles together with a valid finite rectangle.

target_c_selection_circularity.py. Checks the A_n/B_m relation law and the incompatible inequalities behind the failed tail-stabilize/add-witness fixed point. The original attack packet is included unmodified for context. The package README records commands, expected verdicts, file hashes, and the distinction between paper proofs, finite executable checks, and the supplied Lean-certified background.

B Source basis and formal-status note

This report is a response to the supplied files `ATTACK_PROMPT.md`, `CERTIFICATE.md`, and the probes in `attack_mixed_quadrant.zip`. It does not claim to have inspected or recompiled the approximately 41,400-line Lean development. Statements described as “certified” are inherited from the packet’s report; new findings in this document are paper-level arguments supported, where indicated, by finite executable checks. The finite checks do not substitute for Lean proofs of the general theorems.